

Integers, Primes and Roots — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- A **prime** has exactly two factors. **1 is not prime.**
- For **HCF** take the primes in both lists at the lower power; for **LCM** take every prime at the higher power. As a check, $\text{HCF} \times \text{LCM}$ equals the product of the two numbers.
- Where a question says “in index form”, write $2^3 \times 3^2$, not $2 \times 2 \times 2 \times 3 \times 3$.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – TAKING NUMBERS APART

- 1.** Complete the table. Each empty box is worth one mark. For the last column write YES or NO, and you should be able to decide it from the index form alone. [6]

Number	As a product of primes, in index form	A square number?
72		
100		
96		

A number is a perfect square exactly when every index in its prime factorisation is even.

- 2.** Answer both parts.
- (a)** List all the factors of 40. [1]

- (b)** Exactly one of 39, 41, 49 and 51 is prime. Which one, and why are the other three not prime? [1]

- 3.** Write each number as a product of its prime factors, in index form.

(a) 126 [1]

(b) 189 [1]

SECTION 2 – NEGATIVES, POWERS AND ROOTS

4. Work out:

(a) -9×6 [1]

(b) $-72 \div 8$ [1]

(c) $-2 \times -5 \times -4$ [1]

(d) $(-6)^2$ [1]

5. Work out:

(a) $(-2)^4$ [1]

(b) $(-2)^5$ [1]

6. Work out:

(a) $\sqrt{225}$ [1]

(b) the cube root of 343 [1]

(c) 14^2 [1]

(d) 4^3 [1]

SECTION 3 – USING IT

- 7.** A tailor has two rolls of ribbon, one 45 cm long and one 60 cm long. She cuts them into equal pieces, as long as possible, with nothing left over from either roll. Find the length of each piece and the total number of pieces. [3]

- 8.** Two cyclists start together at the same point on a circular track. One completes a lap every 40 seconds, the other every 60 seconds. After how many seconds are they next together at the starting point, and how many laps has each completed by then? [3]

- 9.** Without using a calculator, state which two consecutive whole numbers each of these lies between. You must show the two square or cube numbers you used.

(a) $\sqrt{40}$ [1]

(b) $\sqrt{150}$

[1]

(c) the cube root of 30

[1]

(d) the cube root of 200

[1]

End of paper. Check every answer has a unit or a form the question actually asked for.